

CHAROTAR UNIVERSITY OF SCIENCE AND TECHNOLOGY

First Semester of M.Tech. (CE) Examination

May 2011

CE701 Operating System Design and Concepts

Date: 09/05/2011

Time: 10:00 a.m. to 01:00 p.m.

Maximum Marks: 70

Instructions:

1. Figure to the right indicates full marks
2. Make suitable assumption whenever necessary and mention it clearly
3. Parts of same question should be answered together and in the same sequence.
4. Answers to the sections must be written separately.

SECTION-I

Q-1

- | | |
|--|---|
| (a) Briefly explain the differences between user mode and kernel mode operation. Why two modes of operations are required in UNIX? | 4 |
| (b) Compare C-thread of Mac and P-thread of POSIX in detail. | 4 |
| (c) What are the contents stored in process image? | 3 |

Q-2

- | | |
|--|---|
| (a) Explain in detail about PCB and how process represented in system memory. | 4 |
| (b) Demonstrate the creation of child process and synchronization between parent and child process | 5 |
| (c) Explain opendir() and readdir() function to access directory. | 3 |

OR

- | | |
|---|---|
| (b) Why is it difficult to protect a system in which users are allowed to do their own I/O. | 4 |
| (c) Write the steps followed by the kernel for releasing a buffer. | 4 |

Q-3 Answer the following questions. (Any Three) 12

- | | |
|---|--|
| (a) Compare and contrast the handling of system calls and of exceptions. What are the similarities and differences? | |
| (b) Explain Unlink race condition. | |
| (c) Explain the purpose of open () and close () operations. | |
| (d) Explain the file protection mechanism used in Unix operating system. | |

SECTION-II

Q-4

- | | |
|---|---|
| (a) When buffer is released where it is put in the free list? Justify your answer | 2 |
| (b) How does the kernel differentiate device files and ordinary files? | 2 |
| (c) What are the elements of structure of Message queue? | 2 |
| (d) Enlist the state information are to be maintained in each user thread. | 2 |
| (e) Write multiprocessor algorithm for Wait/Exit. | 3 |

Q-5

- (a) Explain System V Shared memory system calls. 6
- (b) Draw the architecture of Satellite processor distributed system. How to handle signals on Satellite system. 6

OR

- (b) Explain the structure of inode with suitable example. 6

Q-6 Answer the following questions. (Any three) 12

- (a) List and explain major sources of signal generation.
- (b) Explain system V IPC.
- (c) Explain data structures after mount.
- (d) When searching for a block in buffer cache, kernel does not find the block. What does it do.